
What Survives a Rename: Dissociating Lexical and Numeric Robustness in LLM Reasoning

Anonymous Author(s)

Affiliation

Address

email

Abstract

Equal benchmark accuracy can be reached by procedures that behave very differently when the problem’s surface is changed. We probe six frontier and open-weight models on 219 problems spanning arithmetic, classical planning, and dynamic-programming optimisation, presenting each problem in six answer-preserving surface forms and adding a cross-session plan-versus-execution probe and a mid-solve false-state injection probe. Five results emerge. (i) Robustness separates by *what* is changed, not how much: regenerating a problem with new numeric values costs almost nothing (Gemini .909 $\rightarrow$.958) while moving the same problem into a different cover story costs a great deal (.909 $\rightarrow$.523), on the same algorithm at the same difficulty. (ii) Rewriting a problem in formal notation, with entities left intact, is often *harder* than renaming: even the reasoning-trained o4-mini loses 16 points. (iii) Inverting a planning problem’s initial and goal states makes it markedly easier for most models (Claude .172 $\rightarrow$.873). (iv) Fragility has no item-level locus: no problem fails for all models in arithmetic or optimisation, and which surface transform is hardest is a property of the task family (Kendall $W = .42-.68$ across models within a family) and not of the model (Claude $W = .165$, $p = .82$ across families). (v) Robustness is a measurable capacity distinct from accuracy: two methodologically independent probes agree ($r = .243$, $p = .024$, $n = 272$) while neither tracks canonical accuracy. We further show that models absorb injected false intermediate states at rates up to 100% yet recover the correct final answer at near-baseline rates, so compliance and correctness are empirically separable.

1 Introduction

A benchmark score is ambiguous about process. A model that answers a word problem correctly may have executed a general procedure, or may have matched the problem against something structurally similar in its training distribution. Both routes produce the same number, so accuracy alone cannot distinguish them. That ambiguity is not academic for teams shipping kernels, quantised checkpoints, prompt templates, or user-facing features: the same leaderboard score can hide different failure modes once inputs are paraphrased, formalised, or tied to different surface identities in deployment. The instrument is built to make those gaps measurable before they become production incidents, not to replace accuracy.

The natural experiment is to change the problem’s surface while preserving its solution and observe what survives. Prior work establishes that group-level accuracy declines under such perturbation in arithmetic [Mirzadeh et al., 2025, Li et al., 2024], in competition mathematics [Huang et al., 2025,

Hao et al., 2025], across counterfactual task variants [Wu et al., 2024], and in classical planning under predicate obfuscation [Valmeekam et al., 2023].

We ask a finer question: *which* surface properties carry the load, and is the answer a property of the model or of the task? To do that we apply six distinct answer-preserving transformations to the same problems, across three domains that differ in algorithmic character, and we pair the surface probe with two process-level probes on the same instances.

Contributions.

- A dissociation between numeric and cover-story robustness: regenerated parameter values are nearly free, cover-story isomorphs are expensive, on identical algorithms.
- The observation that formal-notation rewriting, which changes no entity names, is frequently a harder perturbation than renaming.
- A direction-of-reasoning effect in planning: goal/initial inversion raises accuracy substantially for most models.
- Evidence that surface fragility is a model–item interaction with no item-level locus, and that the difficulty ordering over transformations is a family-level rather than model-level property.
- Convergent and discriminant validity for robustness as a capacity distinct from benchmark accuracy, using two probes that share no measurement machinery.
- A separation of compliance from correctness under mid-solve false-state injection.
- A practitioner-facing mapping from common shipping decisions (kernel promotion, compression, training mix, prompting, architecture edits, and fairness-style audits on verifiable tasks) to surface probes and to the findings that justify looking beyond Acc_{can} alone (public site section “Why it matters outside the paper”).

2 Related Work

Surface perturbation. Mirzadeh et al. [2025] regenerate GSM8K templates with new names and values and report group-level decline. Their *Vary Name* condition changes proper names only; our W3b changes the entire cover story, and our W6 changes values within a fixed template. The two manipulations are therefore not comparable one to one, and we report the contrast explicitly rather than treating our W3 as a replication of theirs. Valmeekam et al. [2023] introduce obfuscated Blocksworld, replacing domain predicates with arbitrary tokens; our planning rename follows that design. Wang and Zhao [2024] apply nine perturbation types across fifteen datasets and Wu et al. [2024] evaluate eleven counterfactual tasks. These establish that perturbation costs accuracy. We ask which perturbation, on which family, and whether the ordering is stable.

Counterfactual bias benchmarks [Parrish et al., 2022, Sahoo et al., 2024] and steering-based debiasing [Siddique et al., 2025] address attribute sensitivity and repair; our planned W8 transform instead asks whether attribute-swap cost on verifiable-gold items is the same construct as W3a (nonce rename) fragility under the surface-invariance instrument.

Direction in planning. Direction asymmetry in planning has been reported before and linked to search complexity [Ren et al., 2024]. We replicate it on PlanBench Blocksworld items and leave its mechanism open; a pre-registered discriminating test (backward-reasoning instruction on canonical items versus constraint asymmetry per item) is specified in our repository.

Chain-of-thought faithfulness. Turpin et al. [2023] and Lanham et al. [2023] intervene within a single reasoning trace. Our plan-execution probe instead compares a declaration produced in one session against execution in a second session with no shared context, isolating behavioural consistency from within-context self-consistency.

Exposure and frequency. Razeghi et al. [2022] show pretraining term frequency predicts arithmetic accuracy. For closed-weight models corpus membership is not recoverable, so we treat Infinitigram overlap [Liu et al., 2024] as an explicitly labelled public-corpus proximity proxy, used only within family.

3 Method

Problem families. Three domains, chosen for contrasting algorithmic character. **GSM**: 44 arithmetic word problems sampled from GSM8K [Cobbe et al., 2021] across proximity quartiles. **BW**: 65 Blocksworld planning instances [Valmeekam et al., 2023], 50 standard and 15 obfuscated, with a formal PDDL encoding. **ALGO**: 110 dynamic-programming problems in three subtypes that share algorithmic structure but differ by orders of magnitude in public-corpus prevalence: coin change (very common in tutorial text), shortest path (moderate), weighted interval scheduling (rare).

Surface transformations. Each problem is expanded to six variants by deterministic scripts, each preserving the answer exactly except where noted: W1 paraphrase; W2 structural reformatting (adjacency matrix, interval table, equation set); W3 *cover-story isomorph*: entities, actions and framing are mapped into a different real-world domain (e.g. coins to stamps, a phone tariff to a calorie budget, block moves to a reporting chain) while numbers, structure and gold answer are preserved; the mapping is persisted and consumed by the verifier; W4 formal notation ($\LaTeX$ for arithmetic, formal predicates for planning, edge-list notation for graphs); W5 direction reversal (initial and goal states exchanged for planning, source and destination exchanged for graphs, forward-to-backward inversion for arithmetic; undefined for coin change and interval scheduling); W6 procedural regeneration with the same algorithm and parameter ranges but new numeric values, so the gold answer changes while the procedure does not.

Instrument. The variant generator persists the entity mapping it applies, and the domain verifier consumes that mapping, so a correct answer expressed in the renamed vocabulary is recognised as correct. Verification is semantic where the domain permits: a PDDL state simulator checks goal reachability for planning, an exact optimal solver checks optimisation answers, and numeric matching checks arithmetic. Before any model is queried, each item’s gold answer is submitted to its own verifier as a validity check, and each variant is checked to differ from its canonical in the intended way; items failing either check are recorded with a reason and excluded rather than scored.

Process probes. *Plan–execution consistency* (CCI): the model declares a complete step-by-step plan in one session, then executes it step by step in a second session that receives the problem and current state but no plan and no chat history. CCI is the fraction of declared numeric steps matched within $\epsilon = 0.01$. *False-state injection*: identical to the execution phase except that at a sampled interior step the true intermediate value is replaced with a plausible but incorrect one; we classify the model’s response at the injection step and measure final-answer accuracy against the uninjected run.

Models and sampling. Five primary models at $T = 0$, zero-shot chain-of-thought through a single gateway: Claude Sonnet 4, GPT-4o, Gemini 2.5 Flash, Llama-3.1-8B-Instruct [Dubey et al., 2024], and o4-mini; plus DeepSeek-R1-distill-Llama-70B [Guo et al., 2025] on planning only (Probe 1 Blocksworld; $n=6$ models on BW, $n=5$ on GSM and ALGO). Approximately 20,000 model calls.

Statistics. Because procedurally generated banks contain families of near-duplicate items with identical gold answers, ALGO intervals are 10,000-draw percentile bootstraps resampling those families rather than individual items; the 110-item bank comprises 51 such families. GSM and BW use Wilson intervals. Retention $R_{W3} = \text{Acc}_{W3} / \text{Acc}_{\text{can}}$ is reported only where canonical accuracy is at least 0.30, a threshold fixed before analysis.

4 Lexical and Numeric Robustness Dissociate

All results in this section are exploratory with respect to the pre-registration, which governs subsequent calibration and mechanism phases; thresholds reported here (e.g. the $\text{Acc}_{\text{can}} \geq .30$ retention floor) were fixed before the corresponding analyses were run.

The sharpest contrast in our data is between W6 and W3. Both replace content in the problem; W6 replaces the numbers and W3 replaces the cover story. The underlying algorithm and the difficulty class are held fixed in both.

Table 1: Arithmetic per-variant accuracy ($n=5$ primary models). W6 regenerates numeric values; W3 is a cover-story isomorph. Numeric novelty is nearly free; cover-story change is expensive, and the cost varies sharply by model.

Model	Canonical	W1	W2	W3	W4	W5	W6
Claude Sonnet 4	.841	.841	.773	.750	.636	.818	.750
Gemini 2.5 Flash	.909	.818	.636	.523	.477	.614	.958
o4-mini	.841	.864	.818	.841	.682	.886	.833

Table 2: Blocksworld per-variant accuracy ($n=5$ primary models shown; DeepSeek-R1-distill on planning is additional). W5 exchanges initial and goal states.

Model	Canonical	W1	W2	W3	W4	W5	W6
Claude Sonnet 4	.172	.138	.305	.100	.119	.873	.673
GPT-4o	.063	.172	.102	.100	.119	.291	.250
Gemini 2.5 Flash	.391	.259	.237	.133	.068	.764	.404
o4-mini	.781	.862	.831	.483	.475	.909	1.000
Llama-3.1-8B	.016	.052	.017	.050	.000	.000	.019

Gemini answers 90.9% of canonical items correctly, 95.8% when the numbers are regenerated, and 52.3% when the entities are renamed. The procedure evidently generalises across numeric instantiations of the same problem while remaining sensitive to the lexical frame in which those numbers are embedded. Claude shows the same pattern more weakly (.750 on both W6 and W3 against .841 canonical) and o4-mini is essentially flat across both.

Two consequences follow. First, a single scalar “robustness” summarises very little: the same model can be perfectly robust to one class of change and fragile to another. Second, the dissociation localises what the sensitive representation is keyed to. Whatever supports the correct answer is stable under substitution of the quantities and unstable under substitution of the names.

5 Formal Notation Costs More Than Renaming

W4 rewrites the problem in formal notation while leaving entity names untouched. It is consistently among the most damaging transformations, frequently exceeding W3.

On arithmetic, Gemini falls to .477 under W4 against .523 under W3 and .909 canonical; Claude to .636; and o4-mini, which is otherwise near-invariant, to .682 from .841, a 16-point loss that no other transformation inflicts on it. On planning the pattern is stronger still: Claude .119 and Gemini .068 under W4, both below their W3 values.

This is not a lexical effect, since the entities are unchanged. It indicates sensitivity to the representational register of the problem statement independent of its content vocabulary, which the perturbation literature’s focus on renaming does not capture.

6 Direction of Reasoning in Planning

W5 on Blocksworld exchanges the initial and goal states. The transformed problem is a valid planning instance of comparable structure, re-solved with an admissible planner to obtain its gold plan. For most models it is markedly *easier* than the original.

Direction asymmetry in planning has been reported before and linked to search complexity [Ren et al., 2024]. We replicate it on PlanBench Blocksworld items and leave its mechanism open; a pre-registered discriminating test (backward-reasoning instruction on canonical items versus constraint asymmetry per item) is specified in our repository.

Claude moves from .172 to .873, Gemini from .391 to .764, o4-mini from .781 to .909. The effect is large, appears in three of five models, and is absent in the two models at the accuracy floor.

We do not have a mechanism for this. Two hypotheses are testable with the present design: that goal states in the released instances are structurally more constrained than initial states, so exchanging

Table 3: Plan–execution consistency against rename survival, point-biserial with cluster bootstrap (5,000 resamples, seed 42).

Scope	r	95% CI	p	n
Arithmetic	.156	[−.006, .308]	.060	128
Optimisation	.359	[−.043, .664]	.085	144
Pooled	.243	[.027, .414]	.024	272
Pooled, canonical-correct	.256	[.043, .416]	.020	178

them reduces the effective branching factor; or that forward planning from a sparse initial state requires longer unguided search than backward chaining from a specified goal. Distinguishing them requires generating instances with matched initial and goal complexity, which is a small experiment.

7 Fragility Is a Model–Item Interaction

No item-level locus. Among items evaluated on all five primary models, the number failing for every model is 0/110 on the optimisation family and 0/20 on arithmetic. On planning it is 14/64, and all fourteen lie in the obfuscated condition. Outside obfuscated planning there is essentially no universally fragile item. Which problems break under renaming depends on the model, not on an intrinsic property of the problem.

Within-model structure. The contingency coefficient ϕ between canonical and renamed correctness is positive in most cells and varies widely: on optimisation, Claude .68 [.53, .84], GPT-4o .52 [.33, .70], Gemini .33 [.19, .49]; on arithmetic, o4-mini .66 against Gemini .01 and GPT-4o −.03. A high ϕ means the renamed problem set is roughly the canonical set with the hardest items removed; a ϕ near zero means renaming produces a nearly orthogonal set of solvable problems. Models differ in which regime they occupy.

Transformation difficulty is a family property. Ranking W1–W6 by accuracy, the ordering is significantly concordant across models within a family: Kendall $W = .419$ ($p = .049$) on optimisation, .679 ($p < .001$) on planning, .660 ($p = .042$) on arithmetic. Across families within a model it is not: Claude $W = .165$ ($p = .824$), Gemini .238 ($p = .673$). “Which surface transform is hardest” is therefore a property of the task family, and statements of the form “model X is rename-fragile” are not well formed without a family index.

Reliable dissociations occur in arithmetic only. Testing pairwise model–item crossovers against an additive item-plus-model null, arithmetic yields 20 reliable crossovers against a null expectation of 0.57 ($p = .001$), while optimisation and planning are at or below chance. Only in arithmetic is there behavioural evidence for genuinely separable competencies rather than a single ability axis with noise.

Variance loads on different facets by family. A crossed variance decomposition of binary accuracy places .327 of optimisation variance on the model facet against .018 on items ($G_{\text{item}} = .415$), and inverts on arithmetic: items .186, model .009, $G_{\text{item}} = .680$. Arithmetic items differ meaningfully from one another; optimisation items largely do not, and what varies is the model. The two families consequently require different experimental designs, and fitting a latent-strategy mixture confirms this: on arithmetic, BIC selects a single class over $K \in \{1, \dots, 4\}$, so no discrete strategy subpopulations are recovered at this sample size, while optimisation and planning fall below the item-variance threshold for such a model to be identifiable at all.

8 Robustness Is a Capacity Distinct From Accuracy

Surface invariance and procedural consistency are measured here by probes that share no machinery: one compares accuracy across rewritten problem statements, the other compares a declared plan against its execution in an isolated session.

The two probes agree ($r = .243$, $p = .024$), and neither tracks canonical accuracy: the rank correlation between accuracy and rename retention is $.136$ ($p = .642$), and between accuracy and ϕ it is $-.426$ ($p = .146$). Together these give convergent and discriminant validity for robustness as a distinct capacity.

The effect is not detectable within either family alone. At the per-family sample sizes typical of this literature, a real association of this magnitude is invisible, which bears on how such studies should be powered.

Mean consistency by model. On arithmetic, Gemini attains the highest plan–execution consistency (.270), followed by Claude (.231), o4-mini (.220), Llama (.167) and GPT-4o (.108). On optimisation, Claude leads at .384, GPT-4o .317, Llama .214. High consistency does not imply high accuracy: GPT-4o has the lowest arithmetic consistency and a median of zero, yet answers correctly at rates comparable to models with twice its consistency.

9 Compliance and Correctness Are Separable

At a sampled interior step we replace the true intermediate value with a plausible incorrect one. Acceptance is high: Claude 88.5%, GPT-4o 93.4%, o4-mini 100%, Llama 39.3%. Gemini is the outlier, treating every injection as malformed input and proceeding without it (0% compliance).

Final-answer accuracy is nonetheless close to the uninjected baseline: Claude .525 against .541, GPT-4o .508 against .557, o4-mini .377 against .409. The injected value propagates into many subsequent steps, yet the final answer survives.

Models therefore adopt a false premise, carry it forward visibly, and still reach the correct result at near-baseline rates. Any monitoring scheme that treats the intermediate trace as a faithful record of the computation producing the answer must account for this.

A signature of reasoning training. Among rename errors we score *intrusion*: answers that reuse a token from the original surface form after it has been renamed. On optimisation, o4-mini intrudes at .619 (13 of 21 errors), against Claude .250, Gemini .143, GPT-4o .054 and Llama .000. On planning the rate is zero for every model. The reasoning-trained model is distinctively prone to surfacing the pre-rename entity, which is a behavioural signature worth mechanistic follow-up rather than a claim in itself.

10 Programme and Next Experiments

This work establishes the behavioural layer of a staged programme. Each next stage has a stated prediction and decision criterion.

Mechanism. If lexical and numeric robustness dissociate behaviourally (Section 4), the representational correlate should be a rename-induced shift in the accessibility of the answer token that predicts per-item rename survival within a model. Our open-weight pilot cannot test this: on the current item set the available open models solve too few renamed items (1/60 and 0/61) to support a within-model correlation. The experiment requires either easier items or stronger open backbones, run unquantised so that cross-model contrasts are not confounded by precision.

Direction of reasoning. Generating Blocksworld instances with matched initial- and goal-state complexity distinguishes the branching-factor and search-direction accounts of Section 6. Under 100 inference calls.

Exposure. Corpus membership is not recoverable for closed-weight models, so exposure claims require open-corpus models. The planned calibration fits behavioural indicators against known membership on such a model, with the pre-registered criterion that indicators must track exposure at $AUC \geq 0.75$ to be retained.

Training conditions and development. Two further stages follow: characterising the training-data conditions under which invariance forms, which requires controlled training runs; and tracking when invariance emerges relative to accuracy across public checkpoint series.

Bias as surface binding. Under the planned Track B framing, protected attributes in verifiable-gold items are treated as answer-irrelevant surfaces. The planned W8 transform would test whether attribute-swap cost covaries with W3a (nonce rename) cost and whether both localise to a shared representational direction; if so, bias auditing would become a special case of the instrument. No Track B runs have been performed.

11 Limitations

The instrument is designed for per-instance diagnosis, but we do not issue per-instance retrieval or computation labels: across a 270-configuration threshold sweep the retrieval-consistent count ranges from 0 to 170 of 440, and mixture Rasch models select a single class on arithmetic. Per-instance verdicts may also be sensitive to the inference stack (batch size, kernel, accumulation precision) even at temperature zero; we quantify this floor on open-weight models in follow-up work.

12 Conclusion

Perturbing a problem’s surface reveals structure that accuracy hides, but the structure is finer than a single drop. Numeric regeneration is nearly free while entity renaming is expensive on the same algorithm; formal-notation rewriting, which changes no names, is often costlier than renaming; and inverting a planning problem’s direction makes it easier for most models. Which transformation is hardest is a property of the task family rather than the model, and no item is universally fragile outside obfuscated planning. Two independent probes converge on a robustness capacity that is uncorrelated with benchmark accuracy. Finally, models accept injected false premises at rates up to 100% and still recover the correct answer, separating compliance from correctness. Reporting a single accuracy number, or a single robustness number, discards all of it.

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